Homework 17: Kernel and Image of a Linear Transformation

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BCS A, E, F, G

Definition 1. ***Image of a function.*** Recall from single variable calculus that the *image* of a function consists of all the values the function takes in its target space. If f is a function from X to Y $(f: X \to Y)$, then

$$\text{image}(f) = \{f(x): x \in X\} = \{b \in Y: b = f(x), for\ some\ x \in X\}.$$

Theorem 1. ***Image of a Linear Transformation.***

The image of a linear transformation $T(\boldsymbol{x}) = A\boldsymbol{x}$ is *the span of the column vectors of A*. We denote the image of T by $\text{im}(T)$ or $\text{im}(A)$.

Definition 2. ***Kernel of a Linear Transformation.***

The kernel of a linear transformation $T(\boldsymbol{x}) = A\boldsymbol{x}$ from $\mathbb{R}^n$ to $\mathbb{R}^m$ consists of all zeros of the transformation, that is, the solutions of the equation $T(\boldsymbol{x}) = A\boldsymbol{x} = \boldsymbol{0}$. In other words, the kernel of T is the solution set of the linear system $A\boldsymbol{x} = \boldsymbol{0}$. We denote the kernel of T by $\ker(T)$ or $\ker(A)$.

Theorem 2. ***When is $\ker(A) = \{\boldsymbol{0}\}$?***

a. Consider an $m \times n$ matrix A. Then $\ker(A) = \{\boldsymbol{0}\}$ if (and only if) $\text{rank}(A) = n$.

b. Consider an $m \times n$ matrix A. If $\ker(A) = \{\boldsymbol{0}\}$, then $n \leq m$. Equivalently, if $n > m$, then there are nonzero vectors in the kernel of A.

c. For a square matrix A, we have $\ker(A) = \{\boldsymbol{0}\}$ if (and only if) A is invertible.

Summary: ***Various Characterizations of Invertible Matrices.***

For an $n \times n$ matrix A, the following statements are equivalent; that is, for a given A, they are either all true or all false.

i. A is invertible.

ii. The linear system $A\boldsymbol{x} = \boldsymbol{b}$ has a unique solution $\boldsymbol{x}$, for all $\boldsymbol{b}$ in $\mathbb{R}^n$.

iii. $\text{rref}(A) = I_n$.

iv. $\text{rank}(A) = n$.

v. $\text{im}(A) = \mathbb{R}^n$.

vi. $\ker(A) = \{\boldsymbol{0}\}$.

Problem 1. Find the kernel and range of the differential operator $D: P_3 \to P_2$ defined by $D\big(p(x)\big) = p'(x)$. Find the rank and the nullity of the linear transformation D. Write rank-nullity theorem.

Problem 2. Let $S: P_1 \to \mathbb{R}$ be the linear transformation defined by $S\big(p(x)\big) = \int_0^1 p(x)dx$. Find the kernel and range of S. Find the rank and the nullity of the linear transformation S. Write rank-nullity theorem.

Problem 3. Let $T: M_{22} \to M_{22}$ be the linear transformation be defined by taking transposes: $T(A) = A^T$. Find the kernel and range of T. Find the rank and the nullity of the linear transformation T. Write rank-nullity theorem.

Definition 3. ***Isomorphisms.*** Suppose V and W are vector spaces over the same field. We say that V and W are isomorphic, denoted by $V \cong W$, if there exists an invertible linear transformation $T: V \to W$ (called an isomorphism from V to W).

Theorem 3. ***Isomorphisms of Finite-Dimensional Vector Spaces.*** Suppose V is an $n-$dimensional vector space over a field F. Then $V \cong F^n$.

Problem 4. Show that the vector spaces $V = span(e^x, xe^x, x^2e^x)$ and $\mathbb{R}^3$ are isomorphic.

Hint. The standard way to show that two spaces are isomorphic is to construct an isomorphism between them. To this end, consider the linear transformation. $T: \mathbb{R}^3 \to V$. You have to define this map and do the rest.

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Good Luck